

Item Types Design

- We distinguish between ‘materials’, ‘resources’, and ‘equipment’ items.
 - **Materials**, like metals or chemicals, have a notion of quality and technological advancement, where certain techniques or facilities might produce/require certain tiers and material types. Many of the 4-6 tiers of each type are likely to require special resources found on exoplanets, along with new technologies. These types, in addition to the tier-less resources, should encompass all upkeep and the requirements of all production and construction.
 - **Resources**, like oxygen or consumer goods, have some simple identity, and the only advancement is in terms of the scale or ease at which they are produced.
 - **Equipment**, like a spacesuit, vehicle, deployable module, or building, is an individual item or construct with capabilities based on stats and type, and not handled in bulk/stacks. Some are formally ‘items’ and can be carried (e.g. a spacesuit), while others are more abstract (e.g. a power plant or type of engineered microbe) and can use similar stat-like representations but are not inventory items.
- The 6 following tiers are considered for materials. Only 1-3 are available at the start of the game.
 1. **Baseline**: iron, copper, wood, stone capable of basic construction, especially ground-based construction within Earth-like conditions - used in modern housing, tools, power infrastructure, etc.
 2. **Modern**: steel, concrete, glass, silicon, etc. for more advanced construction, for bulk construction exposed to harsh conditions - used in standard modern factories, skyscrapers, tanks, digital infrastructure, etc.
 3. **High-end**: titanium, aluminum, carbon fiber, gold, rare earth metals, etc. for resilient ground and space-capable construction - used in high-end modern ‘smart’ buildings, aircraft, spaceships, high-performance computing, etc.
 4. **Post-modern**: nanomaterials, superconductors, etc. - modern research grade but not feasible at scale, likely allowing fusion, quantum computing, etc. if they became widely available
 5. **Future materials**: likely only theoretical or unconsidered at this point.
 6. **Future materials**: likely only theoretical or unconsidered at this point.
- The 7 following types are considered for materials.
 - **Metallic building materials**: Used for structural support and machinery
 1. Iron and other basic widespread metals that can be used for heavy construction
 2. Steel, magnesium, aluminum, etc. used in modern cities
 3. Titanium and advanced alloys used in ‘smart’ construction and modern aerospace
 4. Post-modern - materials only at research stage now
 5. Future materials - likely only theoretical or unconsidered at this point.
 6. Future materials - likely only theoretical or unconsidered at this point.
 - **Nonmetallic building materials**: Non-metals used as structural base
 1. Wood, stone, etc. for basic widespread construction in earth-like conditions
 2. Concrete, glass, ceramics, etc. used at industrial scale
 3. Carbon fiber, aerogel, etc. non-metallic materials used in advanced construction and aerospace
 4. Post-modern - materials only at research stage now
 5. Future materials - likely only theoretical or unconsidered at this point.
 6. Future materials - likely only theoretical or unconsidered at this point.
 - **Advanced metals**: Used for specialized properties like unusual conductivity, corrosion or heat resistance, etc.
 1. Copper and other basic conductive materials for power infrastructure
 2. Silicon and other materials of modern computing chips, bulk solar panels, etc.

3. Gold, silver, and rare earth metals sought after for high-end electronics and industrial applications
 4. Post-modern - materials only at research stage now
 5. Future materials - likely only theoretical or unconsidered at this point.
 6. Future materials - likely only theoretical or unconsidered at this point.
- **Fuels:** Used for energy and vehicles
 1. Coal, wood, and other simple combustibles
 2. Oil, natural gas, etc. capable of at least fueling cars or generators
 3. Hydrazine, jet fuel, rocket fuel, etc. capable of fueling aircraft and spaceships
 4. Post-modern - materials only at research stage now
 5. Future materials - likely only theoretical or unconsidered at this point.
 6. Future materials - likely only theoretical or unconsidered at this point.
 - **Chemicals:** Process reagents and products for manufacturing and other processes. Includes a wide variety of fertilizer, solvents, catalysts, polymers, rubbers, coatings, etc., but not directly biological
 - * Similar 1-6 scale
 - **Biomaterials:** Nutrients, pharmaceuticals, medical supplies, cell culture media, enzymes, engineered tissues, etc., but not directly industrial unless in a biotech context
 - * Similar 1-6 scale
 - **Nuclear materials:** Fissile materials for nuclear power
 - * Similar 1-6 scale
- **Resources:** Oxygen, GHGs, inert gases, toxic gases, water, food, consumer goods, energy
 - Energy is unique in that, while being consumed and produced like a resource, it is not stored or distributed as a physical item.